Gene Expression Final Project

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```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(gridExtra)
```

```
##
## Attaching package: 'gridExtra'
##
## The following object is masked from 'package:dplyr':
##
##     combine
```

```
library(ggpubr)
library(ComplexHeatmap)
```

```
## Loading required package: grid
## ========================================
## ComplexHeatmap version 2.28.0
## Bioconductor page: http://bioconductor.org/packages/ComplexHeatmap/
## Github page: https://github.com/jokergoo/ComplexHeatmap
## Documentation: http://jokergoo.github.io/ComplexHeatmap-reference
##
## If you use it in published research, please cite either one:
## - Gu, Z. Complex Heatmap Visualization. iMeta 2022.
## - Gu, Z. Complex heatmaps reveal patterns and correlations in multidimensional
##     genomic data. Bioinformatics 2016.
##
##
## The new InteractiveComplexHeatmap package can directly export static
## complex heatmaps into an interactive Shiny app with zero effort. Have a try!
```

```
##
## This message can be suppressed by:
##   suppressPackageStartupMessages(library(ComplexHeatmap))
## ========================================
```

```
library(circlize)
```

```
## ========================================
## circlize version 0.4.18
## CRAN page: https://cran.r-project.org/package=circlize
## Github page: https://github.com/jokergoo/circlize
## Documentation: https://jokergoo.github.io/circlize_book/book/
##
## If you use it in published research, please cite:
## Gu, Z. circlize implements and enhances circular visualization
##   in R. Bioinformatics 2014.
##
## This message can be suppressed by:
##   suppressPackageStartupMessages(library(circlize))
## ========================================
```

```
# This project looks at RNA-seq data for selected genes. The gene chosen was
# RBM5, and CFLAR was chosen as the second gene for comparison. Summary
# statistics and some visualizations are used to examine the gene expression
# data. Tissue type is used as a covariate to compare RBM5 expression
# between tumor and normal samples.

setwd("/Users/simoner/Downloads/Foundations_DS")

# loads in the count matrix and metadata
counts <- read.csv("counts.csv", row.names = 1)
metadata <-  read.csv("meta_data.csv", row.names = 1)
```

```
# selects RBM5 as the primary gene, and CFLAR as the second gene
gene1 <- "ENSG00000003756.17"
gene2 <- "ENSG00000003402.20"
```

```
# this extracts the RBM5 count data
rbm5 <- as.numeric(counts[gene1, ])

# generates summary statistics
summary(rbm5)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     671    2658    3664    4044    5116   12672
```

```
sd(rbm5)
```

```
## [1] 1901.176
```

```r
# extracts the CFLAR count data
cflar <- as.numeric(counts[gene2, ])

# generates summary statistics for second gene
summary(cflar)
```

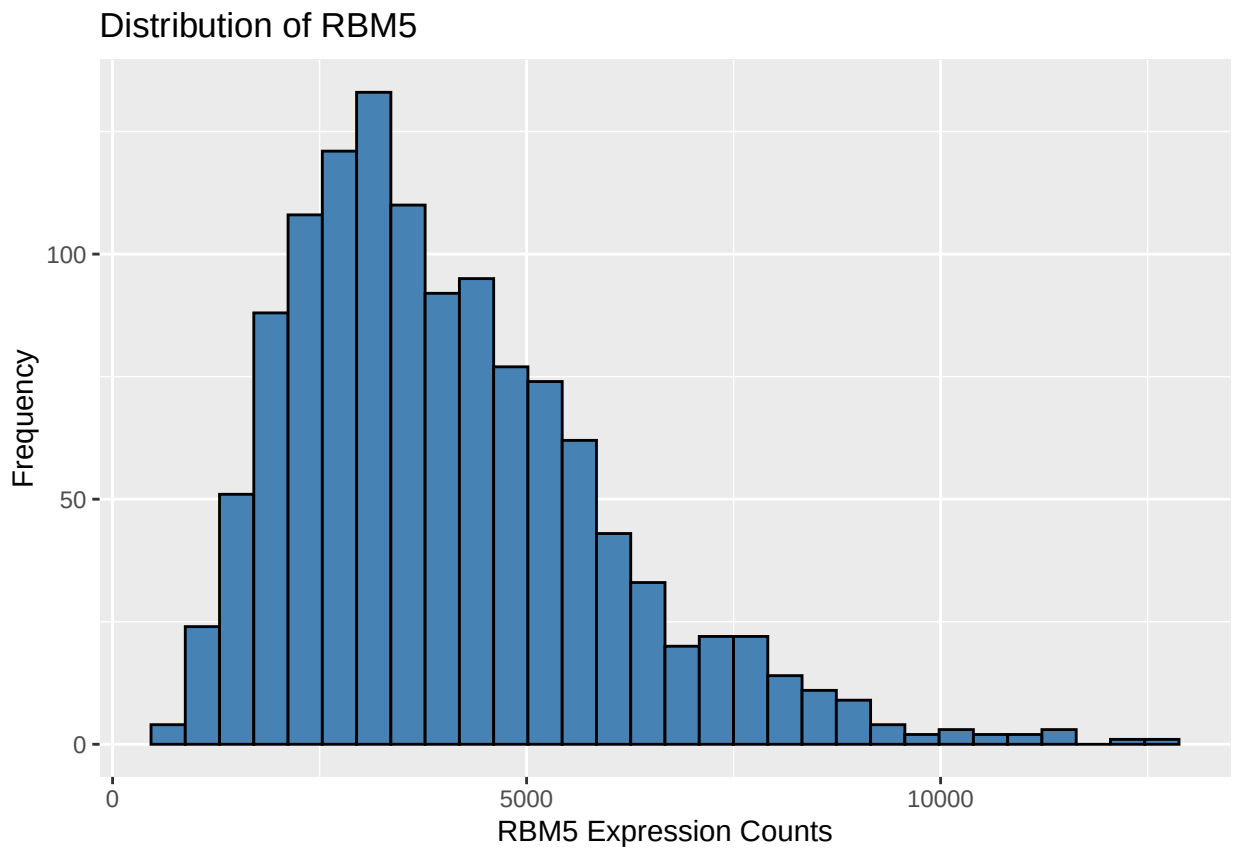
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     664    3346    4799    5442    6720   21182
```

```r
sd(cflar)
```

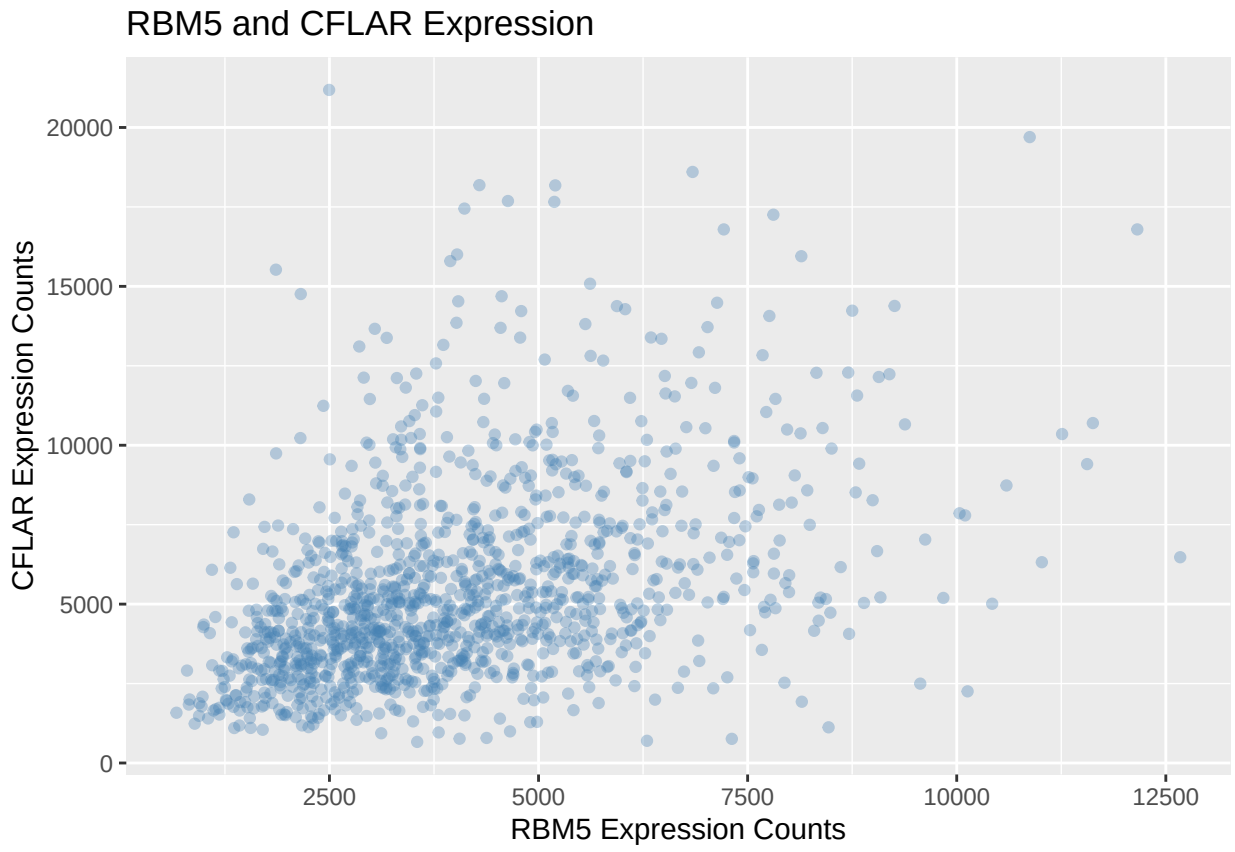
```
## [1] 3040.001
```

```r
# creates a histogram to show distribution of RBM5 counts
ggplot(data.frame(rbm5), aes(x = rbm5)) + geom_histogram(
  fill = "steelblue",
  color = "black") +
  labs(
    title = "Distribution of RBM5",
    x = "RBM5 Expression Counts",
    y = "Frequency"
  )
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



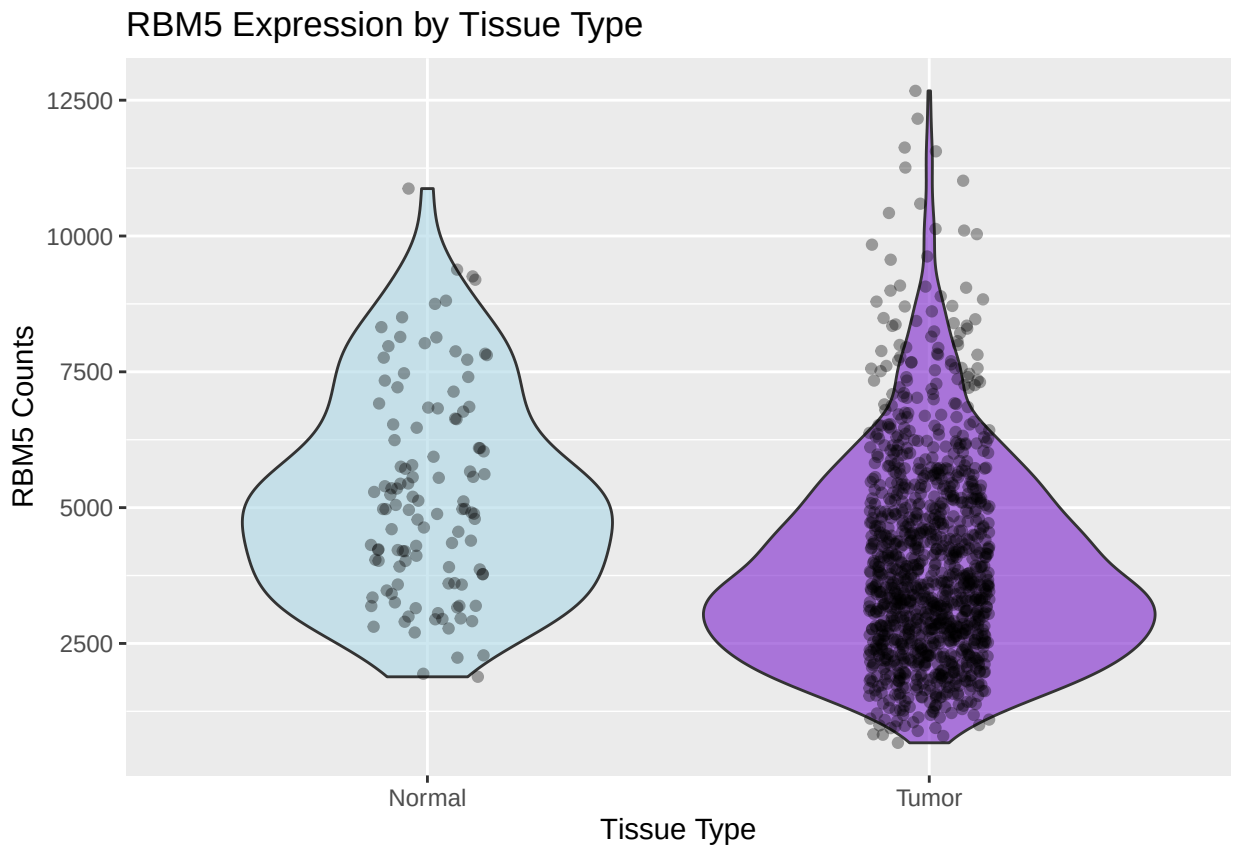
```r
# creates a scatter plot to compare the two genes
ggplot(data.frame(rbm5, cflar), aes(x = rbm5, y = cflar)) +
  geom_point(color = "steelblue",
             alpha = 0.35) +
  labs(
    title = "RBM5 and CFLAR Expression",
    x = "RBM5 Expression Counts",
    y = "CFLAR Expression Counts"
  )
```



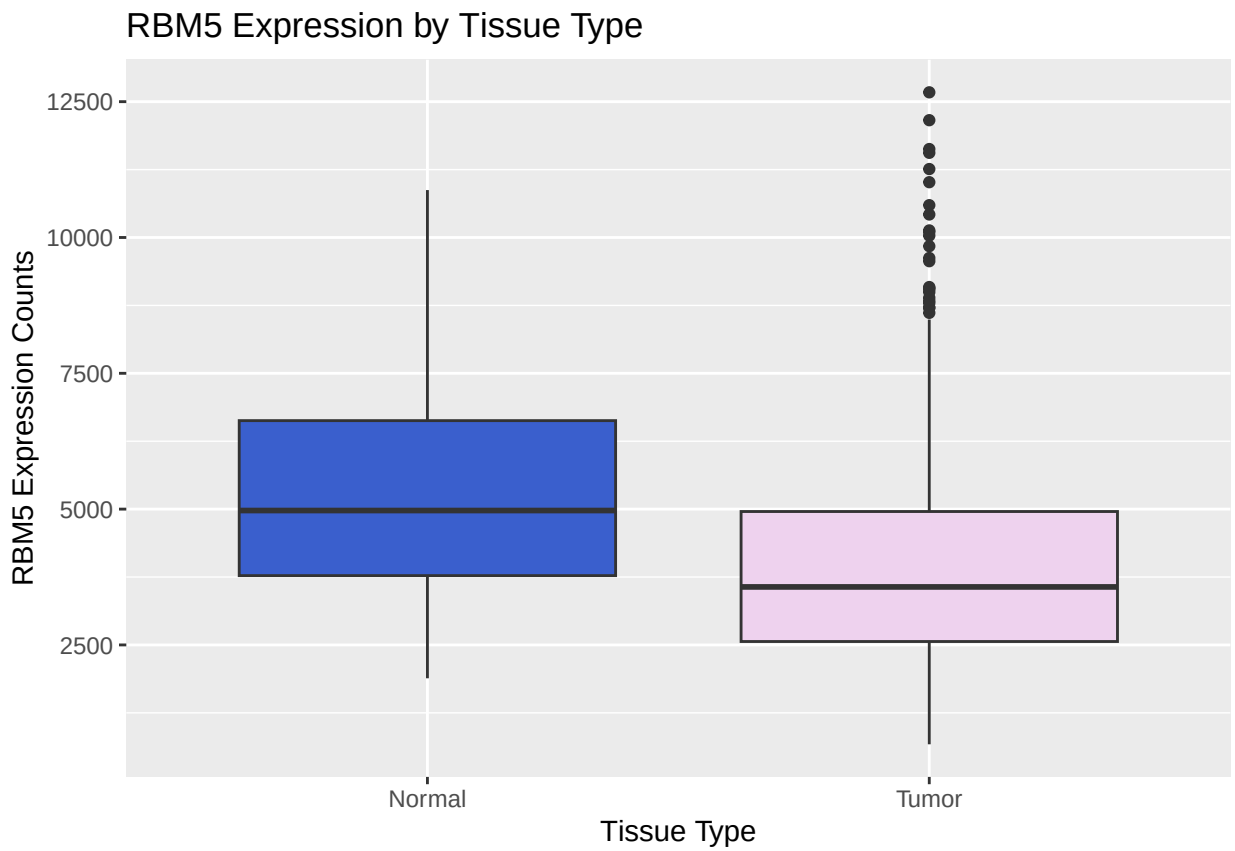

```r
# combines RBM5 counts with tissue type covariate
gene_data <- data.frame(
  rbm5 = rbm5,
  tissue_type = metadata$tissue_type
)

# creates a violin plot to compare RBM5 counts by tissue type
ggplot(gene_data, aes(x = tissue_type, y = rbm5, fill = tissue_type)) +
  geom_violin(alpha = 0.6) + geom_jitter(
    width = 0.12,
    alpha = 0.35,
    size = 1.5
  ) +
  scale_fill_manual(values = c(
    "Tumor" = "purple3",
```

```
    "Normal" = "Lightblue"
  )) +
  labs(
    title = "RBM5 Expression by Tissue Type",
    x = "Tissue Type",
    y = "RBM5 Counts"
  ) + theme(legend.position = "none")
```




```
#creates a box plot to compare RBM5 counts
ggplot(gene_data, aes(x = tissue_type, y = rbm5, fill = tissue_type)) +
  geom_boxplot() +
  scale_fill_manual(values = c(
    "Tumor" = "thistle2",
    "Normal" = "royalblue3"
    )) +
  labs(
    title = "RBM5 Expression by Tissue Type",
    x = "Tissue Type",
    y = "RBM5 Expression Counts"
  ) + theme(legend.position = "none")
```




```
# selects 10 genes and creates smaller matrix with selected genes
heatmap_genes <- rownames(counts)[1:10]

heatmap_genes
```

```
##  [1] "ENSG00000000003.15" "ENSG00000000005.6"  "ENSG00000000419.13"
##  [4] "ENSG00000000457.14" "ENSG00000000460.17" "ENSG00000000938.13"
##  [7] "ENSG00000000971.16" "ENSG00000001036.14" "ENSG00000001084.13"
## [10] "ENSG00000001167.14"
```

```
heatmap_data <- counts[heatmap_genes, ]
```

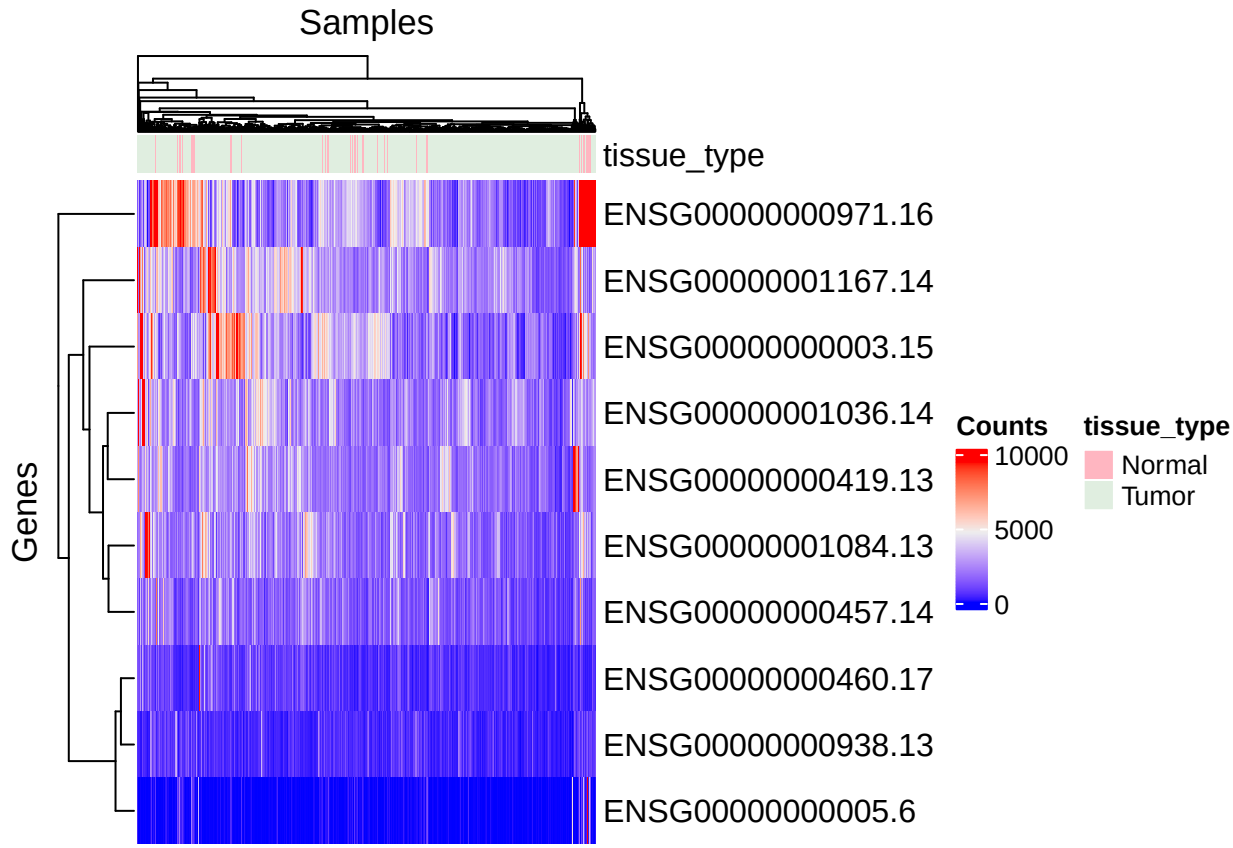
```
# creates an annotation bar showing tissue type
annotation_bar <- HeatmapAnnotation(
  tissue_type = metadata$tissue_type,
  col = list(
      tissue_type = c(
        "Normal" = "lightpink",
        "Tumor" = "honeydew2"
      )
  )
)

# creates a heatmap with the 10 selected genes with tissue type as an
# annotation bar
```

```
Heatmap(
  heatmap_data,
  name = "Counts",
  top_annotation = annotation_bar,
  show_column_names = FALSE,
  row_title = "Genes",
  column_title = "Samples"
)
```

```
## Warning: The input is a data frame-like object, convert it to a matrix.
```

```
## The automatically generated colors map from the 1^st and 99^th of the
## values in the matrix. There are outliers in the matrix whose patterns
## might be hidden by this color mapping. You can manually set the color
## to ‘col‘ argument.
##
## Use ‘suppressMessages()‘ to turn off this message.
```




```
# creates a CDF plot to compare RBM5 expression counts by tissue type
ggplot(gene_data, aes(x = rbm5, color = tissue_type)) +
  stat_ecdf(geom = "step") +
  labs(
    title = "Cumulative Distribution of RBM5 by Tissue Type",
    x = "RBM5 Expression Counts",
```

```
    y = "Cumulative Probability",
    color = "Tissue Type"
  )
```

